

Seasonal Wardrobe Swap SOP (1x Episode: four out, four in, in situ)

One episode swaps four items at the wardrobe bench where they live. The four out-of-season items go first. They are folded, one at a time, and laid in the box. Then the four in-season items come out of the box. Two go on the rail. Two go on the shelf. The episode ends when the box is closed and stowed in its bay. Never split the four out and the four in into two episodes.

The base is **passive**: it has no drive of its own, so it is pushed by hand up to the bench in front of the wardrobe and locked there, and nothing is carried away to another room. Everything the episode touches is already there when recording starts: the bench, the out stack, the open box with the four in-season items in it, the lid beside the box, the empty rail above the left end, the empty shelf above the right end, and the empty stow bay under the shelf.

The wardrobe is worked **as found**. The rail is empty. The shelf is empty. The stow bay is empty. The box stands open on the bench, its lid off and lying flat beside it. The bench top is clear but for the out stack, the box, and the lid. The episode ends with four items folded in the box, two items hanging on the rail, two items lying on the shelf, and the box closed and stowed.

This is an in-situ task, and four things follow from that. First, **everything happens on this bench**: nothing is carried away and nothing new is brought in. Second, **every carry goes along the front lane, low and level**: a thing is lifted just off what it stands on, brought forward into the lane, and carried along the lane close over the bench. The only lift that goes up is the hook going onto the rail. Third, **the fixtures never move**: nothing leans on, pushes, drags, or lifts the bench, the rail, the shelf, or the stow bay. Fourth, **nothing is ever dragged**: every pick lifts straight up off the surface before it moves anywhere.

Clothes go on the bench, in the box, on the rail, or on the shelf, and nowhere else. Nothing ever touches the floor. An item that lands on the floor is a drop. It is not picked back up. It is reported.

The bench is set up in one of three ways. The rail, the shelf, the stow bay, and the fold square never move. Only the out stack, the box, and the lid do.

- **Config L**: the out stack lies at the **front left**. The box stands at the **front right**, with the lid past it at the right end.
- **Config M**: the out stack lies at the **front middle**, just in front of the fold square. The box stands at the **front right**, with the lid past it at the right end.
- **Config R**: the out stack lies at the **front right**. The box stands at the **front left**, with the lid past it at the left end.

One config per episode, chosen before recording and never changed mid-episode. Where a step depends on the setup it says so on an **IF** line. Look at the bench and follow the line that matches.

Same-side rule: the gripper on the out stack's side takes each out item to the fold square, and the gripper on the box's side lifts each hang item out of the box. In Config M the out stack lies in the shared middle and the right gripper takes it. **Hand-over rule:** in Config L and M the rail is too far for the right gripper, so the right gripper lifts each hang item out of the box and hands it over to the left gripper above the middle of the front lane, and the left gripper hangs it. In Config R the box is on the left and the left gripper lifts and hangs each hang item itself. Nothing else is ever passed from one gripper to the other.

Everything else is done by both grippers together and does not change with the config: the folds on the fold square, every flat pack, the lid, and the box are lifted with a gripper on each side, the **left gripper on the left side and the right gripper on the right side**.

The order never changes: **fold all four out items into the box; take the two hang items out and hang them; take the two shelf items out and shelve them; put the lid on; stow the box**. No in-season item comes out of the box until all four folded items are in it.

Setup

Complete the base positioning and all the checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the bench and stop it **square to the front edge of the bench**, so the front edge runs straight across the frame and neither end sits nearer than the other.
2. Stop it **centered on the fold square**, so the rail above the left end and the shelf above the right end are the same distance from the middle of the base.
3. Stop it **close enough** that both grippers reach the far edge of the fold square without either arm extending, and **far enough** that no arm, wrist, or part of the base touches the bench front, the rail, or the shelf while both arms work.
4. Check the **height**: with the base parked, the gripper on the out stack's side lifts the top item off the out stack, brings it forward into the front lane, and carries it to the fold square, without the wrist or the forearm touching the rail or the shelf.
5. Check the **rail**: the **left gripper** lifts a hanger hook up onto the rail and lets go, without extending and without touching the shelf or the other arm.

6. Check the **bench**: the gripper on the out stack's side reaches the out stack, the gripper on the box's side reaches the far half of the box, both grippers together reach both sides of the box, the lid, the shelf, and the stow bay, all without extending and without knocking anything over.
7. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
8. If any of lines 1 to 6 fails, push the base to a new park by hand and start again at line 1. Do not work a bench the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the bench, the rail, or the shelf hard enough to shift it, nothing and nobody touches it, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the fold square and its frame includes the whole bench top, the out stack, the fold square, the open box with both its halves, the lid, the rail, the shelf, and the stow bay.
3. The camera reads the bench from the front, so where a thing lies and whether it lies flat and square are readable.
4. The camera reads the inside of the box from the front, so how many items are in each half is readable.
5. The camera reads the rail from the front, so whether a hook sits all the way over it is readable.
6. The camera reads the front edge of the shelf, so whether a folded item hangs over it is readable.
7. Both arms are at home with grippers open.
8. The **left arm** reaches the left half of the fold square, the rail, the left side of the box, the left side of the lid, the left side of the shelf, the left side of the stow bay, and the middle of the front lane, without extending to a joint limit.
9. The **right arm** reaches the right half of the fold square, the right side of the box, the right side of the lid, the right side of the shelf, the right side of the stow bay, and the middle of the front lane, without extending to a joint limit.
10. The gripper on the out stack's side reaches the stack, and the gripper on the box's side reaches the hooks standing in the far half of the box, without extending.
11. Both grippers reach the fold square at the same time, one at each side of a flat item, without the arms touching each other.
12. A gripper lifts a thing just off what it stands on and carries it low and level along the front lane. It never carries one over the rail, over the shelf, or across the back of the bench.

13. The two arms do not collide. An arm working at the fold square stays out of the other arm's path to the box.
14. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 13 hold.

Materials checklist

1. The **bench** is one long flat top set in front of the wardrobe. Nothing moves it, leans on it, or stands anything on its edge.
2. The bench top is dry, clean, and clear but for the out stack, the open box, and the lid.
3. The **fold square** is the clear flat middle of the bench. It is wide enough for one item to lie out flat, with a hand's width of clear bench all around it.
4. The **out stack** lies flat at the spot this episode's config puts it: the **front left in Config L**, the **front middle just in front of the fold square in Config M**, the **front right in Config R**. It holds four out-of-season items, one on top of another, each laid out flat and squared to the one under it.
5. Each of the four **out items** is a soft folding item: a top, a jumper, or a pair of trousers. Nothing is stiff. Nothing has a hood, a hard buckle, or an open zip.
6. Each of the four out items is clean and dry, and it is small enough to fold to a flat pack that fits in the near half of the box with a finger's width to spare on every side.
7. The **storage box** stands open at the **box spot** this episode's config puts it, square to the front edge, with its lid off: the **front right in Config L and M**, the **front left in Config R**.
8. The box has a low **divider** across its middle. The half nearest the front edge is the **near half**. The half nearest the back is the **far half**.
9. The near half is empty at the start. It takes the four folded items in one stack.
10. The far half holds the four **in items** at the start, in one stack, in a fixed order. The two on top are the **hang items**. The two under them are the **shelf items**.
11. Each hang item is folded once over its own hanger, with the **hook** standing up clear of the fold, so a gripper closes on the hook and nothing else.
12. Each shelf item is folded flat, with no hanger in it.
13. The **lid** lies flat on the bench past the box, out of the front lane: at the **right end in Config L and M**, at the **left end in Config R**. It is the right lid for this box and it sits down on all four sides.
14. The **rail** is a short hanging rail above the left end of the bench. It is empty at the start. It is fixed and does not turn or swing. It is low enough for the left gripper to put a hook on it without extending.
15. There is room on the rail for two hangers side by side, with a hand's width between them and a hand's width clear at each end.

16. The **shelf** is an open shelf above the right end of the bench. It is empty, flat, and dry at the start. It is deep enough for a folded item to lie on it with nothing hanging over the front edge.
17. The **stow bay** is the open bay under the shelf at the right end. It is empty at the start. It is wide enough and deep enough for the closed box to sit inside it with nothing sticking out past the front.
18. The **front lane** is the strip of bench along the front edge from one end to the other. Apart from the out stack and the box standing at their spots, it is empty at the start. Nothing is ever put down in it.
19. The floor around the base is clear and dry.
20. Nothing else is on the bench, on the rail, on the shelf, or in the bay.

Workspace layout

Nothing is marked or taped out. Every place below is judged by eye from the bench and the wardrobe themselves.

- **Back row, left to right:**
 - **Rail**, above the left end of the bench. Empty at the start, holds two hangers at the end. Never moved.
 - **Shelf**, above the right end of the bench. Empty at the start, holds two folded items at the end. Never moved.
- **Middle row:**
 - **Fold square**, the clear middle of the bench. Every fold happens here. One item is on it at a time. Nothing is ever left on it.
- **Front row:**
 - **Out stack**: front left in Config L, front middle in Config M, front right in Config R. The four out-of-season items lie here in one flat stack. It empties as the box fills.
 - **Box spot**: front right in Config L and M, front left in Config R. The open box stands here. Its near half takes the four folded items. Its far half holds the four in-season items at the start.
 - **Lid spot**: the end of the bench past the box, out of the front lane: the right end in Config L and M, the left end in Config R. The lid lies flat here until Step 7.
 - **Stow bay**, under the shelf at the right end. Empty at the start. The closed box goes here at the end.
- **Front lane**: the clear strip of bench along the front edge. Every carry goes along it. Nothing is ever put down in it.

No carry passes over the rail, over the shelf, or across the back of the bench, and nothing travels to the far side of the bench.

Arm assignments

- **The gripper on the out stack's side takes each out item to the fold square:** the left gripper in Config L, the right gripper in Config M and R.
- **Both grippers fold.** On the fold square the left gripper folds the left side in, the right gripper folds the right side in, and both together fold the bottom half up. The right gripper flattens every new fold.
- **Both grippers carry every flat pack, the lid, and the box,** the left gripper on the left side and the right gripper on the right side.
- **The gripper on the box's side lifts each hang item out by its hook:** the right gripper in Config L and M, the left gripper in Config R. **The left gripper hangs every hang item on the rail.** In Config L and M the right gripper hands the hang item over to the left gripper above the middle of the front lane.
- Only one thing is held at a time, by one gripper or by both. No grip is changed while an item is off the bench, except the hang-item hand-over in Config L and M. Every item is put down flat and let go under control.

Vocabulary

- **In situ:** the wardrobe is worked where it is. Nothing is carried away to another room, and nothing about the wardrobe is rearranged for the task.
- **Passive base:** the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Config:** which of the three bench layouts this episode uses. It is set before recording and never changes mid-episode.
- **Out item:** one of the four out-of-season items in the out stack. All four are folded and boxed before anything comes out of the box.
- **In item:** one of the four in-season items in the far half of the box. Two hang. Two go on the shelf.
- **Hang item:** an in item folded over its own hanger, with the hook standing up. The two hang items are the two on top of the in stack.
- **Shelf item:** an in item folded flat with no hanger. The two shelf items are the two under the hang items.
- **Hook:** the curved top of a hanger and the short straight neck under it, above the hanger body. It is the only part of a hang item a gripper ever closes on.
- **Top of the stack:** the item lying highest with nothing on it. The top one is always taken. One is never pulled out from under another.
- **Fold square:** the clear middle of the bench. One item lies out flat here while it is folded. It is empty before an item comes and empty after the item leaves.

- **Laid out flat:** the item lies on the fold square face up, spread out, with no part folded under itself, no sleeve trapped, and no bunching.
- **Fold:** one move that brings one part of the item over onto another part and leaves it lying flat. Each item gets three folds and no more.
- **Flatten:** the right gripper's closed face runs once along a new fold, from one end to the other, close over the bench, to press the fold down. It does not push the item along the bench.
- **Square fold:** the folded edges run straight, the corners line up within a finger's width, and no sleeve, hem, or corner sticks out of the pack.
- **Flat pack:** the finished folded item. It lies flat, holds its shape when let go, and fits in the near half of the box with room on every side.
- **Near half:** the half of the box nearest the front edge of the bench. It is empty at the start and holds the four flat packs at the end.
- **Far half:** the half of the box nearest the back. It holds the four in items at the start and is empty at the end.
- **Divider:** the low wall across the middle of the box that keeps the two halves apart. Nothing is ever laid across it.
- **Just off:** lifted only until the thing's lowest part clears what it was lying on, and no higher.
- **Low carry:** the thing is lifted just off, brought forward into the front lane, and carried along the lane close over the bench and level. It never goes over the rail, over the shelf, or across the back of the bench.
- **Front lane:** the clear strip of bench along the front edge. Every carry goes along it. Nothing is put down in it.
- **Hand over:** in Config L and M only. The right gripper holds the hang item still by the curve of its hook above the middle of the front lane, the left gripper closes on the neck of the hook just below it, and then the right gripper opens. The hanger is never let go between the two grips.
- **Hanging right:** the hook sits all the way over the rail, the hanger hangs straight down, the item hangs level on it, and it does not swing or slide along the rail when the gripper lets go.
- **Sitting right on the shelf:** the flat pack lies flat on the shelf, square to the shelf's front edge, fully on the shelf with nothing hanging over any edge, and beside the other pack and not on top of it.
- **Lid on right:** the lid lies down on the box on all four sides, square with the box, with no corner riding up and no gap visible from the front.
- **Stowed right:** the closed box sits flat in the stow bay, square to the bay, fully inside it, with no part sticking out past the front of the bay.
- **Drop:** an item, a hanger, or the lid that falls on the floor. It is not picked back up. It is reported.

- **Loose item:** an item that ends up on the bench outside the out stack, the fold square, the box, the rail, or the shelf. It is left where it lands and cleared in the reset, never on camera.
- **Back and clear:** the arm is drawn back so that no part of it is over the fold square, the box, the front lane, or the bench.

Handling standard

- Move exactly one thing at a time. Nothing is passed from one gripper to the other, except the hang-item hand-over in Config L and M.
- Every lift comes straight up, just off the surface, before the thing moves anywhere.
- Every carry is a low carry along the front lane.
- Nothing is put down anywhere but its own place in the step being done, and never in the front lane.
- Nothing rests on a fixture. No gripper, wrist, or forearm leans on the bench, the rail, the shelf, or the bay, and no push is ever hard enough to move the box on the bench.

Steps

Steps 1, 3, and 4 depend on the config: the gripper on the out stack's side takes each out item, the gripper on the box's side lifts each hang item, and in Config L and M the hang item is handed over to the left gripper. Step 2 repeats Step 1 with the same IF line. Every other step is the same in all three.

Step 1: Fold the first out item and lay it in the box

Goal: the first out item lies as a flat pack in the near half of the box, and the fold square is empty.

1.1 Take the top item to the fold square

- **IF Config L:** the **left gripper** closes on the top item of the out stack at the middle of its near edge.
- **IF Config M or R:** the **right gripper** closes on the top item of the out stack at the middle of its near edge.

Then, in all three:

- Lift it straight up off the stack, just off, and bring it forward into the front lane.
- Carry it along the lane to the fold square in a low carry and lay it down.

Check: one item lies on the fold square and the out stack is one item shorter. The item under it has not come with it and has not slid. If two items came up together, the gripper that lifted lays both back on the stack, squares them, and takes the top one again.

1.2 Lay it out flat

- The **left gripper** and the **right gripper** each pinch the item at its own side and open it out on the fold square, face up, so it lies spread out and flat.
- Each gripper pulls the sleeves and the corners out on its own side, so no part is folded under itself.
- Both grippers square it to the front edge of the bench.

Check: the item lies flat, face up, and square, with nothing folded under it and no sleeve trapped. The whole item is on the fold square. If a part is under itself, the gripper on that side lifts that part, pulls it clear, and lays it flat again.

1.3 Fold it three times

- The **left gripper** folds the left side in to the middle. The **right gripper** flattens the new fold.
- The **right gripper** folds the right side in to the middle. The **right gripper** flattens the new fold.
- The **left gripper** and the **right gripper** each pinch a near corner and fold the bottom half up onto the top half. The **right gripper** flattens the new fold.

Check: the pack is a square fold. Its edges run straight, its corners line up, and no sleeve or corner sticks out. It is three folds, not two and not four. If a corner sticks out, the gripper on that side opens the last fold, lines the edge up, and folds it again. Make no more than two extra tries on one item.

1.4 Lay it in the near half

- The **left gripper** closes on the left side of the flat pack and the **right gripper** on the right side, and both lift it straight up off the fold square, just off, so it keeps its shape.
- Carry it along the front lane to the box in a low carry. Keep it level the whole way.
- Lower it straight down into the near half of the box, flat and square, and open both grippers.

Check: the pack lies flat in the near half. It is fully inside that half, it does not lie across the divider, and it has not fallen open. The far half has not been touched. If it lies across the divider, both grippers lift it straight up, move it over, and lower it again.

Expected state: one flat pack lies in the near half of the box. The out stack holds three items. The fold square is empty. The far half still holds all four in items. Both grippers are back and clear.

Step 2: Fold the other three out items the same way

Goal: all four out items lie as flat packs in the near half of the box, and the out stack is empty.

2.1 Do the same for the second item

- Do 1.1, 1.2, 1.3, and 1.4 again, with the same IF line in 1.1, for the item now on top of the out stack.
- Lay the pack flat in the near half, on top of the first pack, square with it.

Check: two packs lie in the near half, one on the other, square, and both fully inside that half. The out stack holds two items.

2.2 Do the same for the third item

- Do 1.1, 1.2, 1.3, and 1.4 again for the item now on top of the out stack.
- Lay the pack flat in the near half, on top of the second pack, square with it.

Check: three packs lie in the near half, one on another, square, and all fully inside that half. The out stack holds one item.

2.3 Do the same for the fourth item

- Do 1.1, 1.2, 1.3, and 1.4 again for the last item on the out stack.
- Lay the pack flat in the near half, on top of the third pack, square with it.

Check: four packs lie in the near half, one on another, square, and all fully inside that half. The out stack spot is empty bench. The stack of four sits below the top of the box, so the lid will go on later. If the stack stands proud of the box, the **right gripper** presses the top pack down once, flat and gentle, without pushing the box.

Expected state: four flat packs lie in the near half of the box. The out stack spot is empty. The fold square is empty. The four in items still lie in the far half. Both grippers are back and clear.

Step 3: Hang the first in item

Goal: the first hang item hangs on the rail, straight and level, and the far half holds three items.

3.1 Lift it out by the hook

- **IF Config L or M:** the **right gripper** closes on the hook of the top item in the far half, on the hook and nothing else, lifts it straight up out of the box until the item hangs clear of the box and the divider, brings it forward into the front lane, and carries it in a low carry to the middle of the lane. There it **hands over:** the **right gripper** holds still, the **left gripper** closes on the neck of the hook just below it, and the **right gripper** opens and comes back and clear.
- **IF Config R:** the **left gripper** closes on the hook of the top item in the far half, on the hook and nothing else, lifts it straight up out of the box until the item hangs clear of the box and the divider, and brings it forward into the front lane.

Then, in all three:

- Do not let the item drag on the box edge. Do not let it swing.

Check: the item hangs from the hook in the **left gripper**, straight and clear of the box. Nothing came up with it. The packs in the near half have not moved. If a second item came up with it, the gripper that lifted lowers both back into the far half, lets go, and takes the top hook again.

3.2 Carry it to the rail

- The **left gripper** carries it along the front lane to the left end of the bench in a low carry, keeping the hook level and the item hanging straight.
- Keep the item clear of the bench top, the box, and the lid. Do not let it brush the shelf.
- Bring it in front of the rail, then lift the hook up to the rail.

3.3 Put the hook on the rail

- The **left gripper** lowers the hook onto the rail, all the way over it, near the left end of the rail.
- Let the hanger settle and hang straight down.
- Open the gripper and draw it back away from the hanger.

Check: the item is hanging right. The hook is all the way over the rail, the hanger hangs straight, the item hangs level, and it does not swing or slide when the gripper lets go. If the hook sits on the edge of the rail, or the item hangs crooked, the **left gripper** closes on the hook again, lifts it just off the rail, and puts it on again.

Expected state: one item hangs on the rail. The far half holds three items. The near half holds four packs. Both grippers are back and clear.

Step 4: Hang the second in item

Goal: the second hang item hangs on the rail beside the first, and the far half holds two items.

4.1 Lift it out by the hook

- **IF Config L or M:** the **right gripper** closes on the hook of the item now on top of the far half, lifts it straight up out of the box until it hangs clear, brings it forward into the front lane, carries it to the middle of the lane, and **hands over** to the **left gripper** the same way as in Step 3.1.
- **IF Config R:** the **left gripper** closes on the hook of the item now on top of the far half, lifts it straight up out of the box until it hangs clear, and brings it forward into the front lane.

Check: the item hangs from the hook in the **left gripper**, straight and clear of the box. Nothing came up with it. If it catches on the box edge, the gripper that lifted lowers it back in, lets go, takes the hook again, and lifts straighter.

4.2 Carry it to the rail

- The **left gripper** carries it along the front lane to the left end of the bench in a low carry, keeping the hook level and the item hanging straight.
- Keep it clear of the item already on the rail. Do not knock it or push it along the rail.

- Bring it in front of the rail, then lift the hook up to the rail.

4.3 Put the hook on the rail

- The **left gripper** lowers the hook onto the rail, all the way over it, a hand's width to the right of the first hanger.
- Let the hanger settle and hang straight down.
- Open the gripper and draw it back away from the hanger.

Check: the item is hanging right, and the two items hang apart and do not touch. The first item has not moved along the rail. If the two touch, the **left gripper** closes on the second hook, lifts it just off the rail, and puts it on again further along.

Expected state: two items hang on the rail, apart and straight. The far half holds two items, both folded flat with no hanger. The near half holds four packs. Both grippers are back and clear.

Step 5: Put the first shelf item on the shelf

Goal: the first shelf item lies flat on the shelf, square and fully on it, and the far half holds one item.

5.1 Lift it out flat

- The **left gripper** closes on the left side of the top item in the far half, at the middle of its side, and the **right gripper** closes on the right side the same way.
- Lift it straight up out of the box, keeping it flat and level, so it does not fall open.
- Do not let it drag on the box edge or on the divider.

Check: the pack comes out flat and holds its shape. Nothing came up with it. If it falls open in the air, both grippers lower it back into the far half, let go, close on it again lower down, and lift it again.

5.2 Carry it to the shelf

- Both grippers carry it along the front lane to the right end of the bench in a low carry, keeping it flat and level the whole way.
- Do not carry it over the box or over the lid.
- Bring it in front of the shelf, then move it straight back, over the shelf.

5.3 Lay it on the shelf

- Both grippers lower it straight down onto the shelf, to the left side of the shelf.
- Lay it flat, square to the shelf's front edge, fully on the shelf.
- Open both grippers and draw them back away from the item.

Check: the item is sitting right on the shelf. It lies flat, square, and fully on, with nothing hanging over any edge, and it does not slide when the grippers let go. If a corner hangs over the front edge, both grippers close on it again, lift it just off the shelf, and lay it down again further back.

Expected state: one folded item lies on the shelf. Two items hang on the rail. The far half holds one item. The near half holds four packs. Both grippers are back and clear.

Step 6: Put the second shelf item on the shelf

Goal: the second shelf item lies flat beside the first, the far half is empty, and the box is ready to close.

6.1 Lift it out flat

- The **left gripper** closes on the left side of the last item in the far half, at the middle of its side, and the **right gripper** closes on the right side the same way.
- Lift it straight up out of the box, keeping it flat and level, so it does not fall open.
- Do not let it drag on the box edge or on the divider.

Check: the pack comes out flat and holds its shape. The far half is now empty, with nothing left in the bottom of it. If something is left behind, both grippers lift it out the same way before going on.

6.2 Carry it to the shelf

- Both grippers carry it along the front lane to the right end of the bench in a low carry, keeping it flat and level the whole way.
- Do not carry it over the box or over the lid.
- Bring it in front of the shelf, then move it straight back, over the shelf.

6.3 Lay it beside the first

- Both grippers lower it straight down onto the shelf, to the right of the first item, a hand's width away from it.
- Lay it flat, square to the shelf's front edge, fully on the shelf.
- Open both grippers and draw them back away from the item.

Check: the item is sitting right on the shelf, beside the first item and not on top of it. The first item has not moved. If the two touch or one sits on the other, both grippers close on the second one, lift it just off the shelf, and lay it down again beside the first.

Expected state: two folded items lie side by side on the shelf. Two items hang on the rail. The far half of the box is empty. The near half holds four packs. Both grippers are back and clear.

Step 7: Close the box

Goal: the lid is on the box, square and down on all four sides.

7.1 Look in the box

- Look into the box from the front.
- Count four flat packs in the near half. Count nothing in the far half.

Check: the near half holds four packs and the far half is empty. If the far half is not empty, do Step 5 or Step 6 again for what is left before closing the box. Never put the lid on a box with an item still in it.

7.2 Pick up the lid

- The **left gripper** closes on the left side of the lid at the middle of its side, and the **right gripper** closes on the right side the same way.
- Lift the lid straight up off the bench, just off.
- Carry it along the front lane to the box in a low carry. Keep it flat and level.

Check: the lid is held flat and level. Nothing came up under it. The box has not moved.

7.3 Put the lid on

- Both grippers bring the lid over the box, lined up with it on all four sides.
- Lower it straight down onto the box until it sits down on all four sides.
- Open both grippers and draw them back away from the lid.

Check: the lid is on right. It is square with the box, down on all four sides, with no corner riding up and no gap visible from the front. If a corner rides up, both grippers close on the lid again, lift it just off, line it up, and lower it again.

Expected state: the box stands closed at the box spot. The bench top is clear but for the box. Both grippers are back and clear.

Step 8: Stow the box and end the episode

Goal: the box sits square in the stow bay, and the wardrobe is left as the episode wants it.

8.1 Lift the closed box

- The **left gripper** closes on the left side of the box and the **right gripper** on the right side, both low down on the box body, below the lid.
- Do not hold it by the lid. Do not hook a gripper under the lid edge.
- Lift the box straight up off the bench, just off.

Check: the box is held level, by its body and not by its lid. The lid has not lifted off. If the lid lifts, both grippers lower the box, put the lid back on, take the body again lower down, and lift again.

8.2 Carry it to the bay

- Both grippers carry the box along the front lane to the right end of the bench in a low carry, keeping it level the whole way.

- Do not carry it over the shelf. Do not knock the items on the shelf.
- Bring it in front of the stow bay.

8.3 Put it down in the bay

- Both grippers lower the box straight down into the stow bay.
- Put it down flat, square to the bay, fully inside it.
- Open both grippers and draw them back away from the box.

Check: the box is stowed right. It sits flat, square, fully inside the bay, with no part sticking out past the front. If it sits crooked or sticks out, both grippers close on the box body again, lift it just off the bay floor, and put it down again.

8.4 End the episode

- Confirm the rail: two items hang on it, straight, level, apart, and not swinging.
- Confirm the shelf: two folded items lie flat on it, side by side, square, fully on the shelf.
- Confirm the box: it is closed, and it sits square and fully inside the stow bay.
- Confirm the bench: the out stack spot, the fold square, the front lane, the box spot, and the lid spot are all empty and clear.
- Confirm the fixtures stand where they started, and the base has not moved: it is still square to the bench and still locked.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the bench is set for the next episode.

1. Take the two items off the rail. Leave the rail empty.
2. Take the two folded items off the shelf. Leave the shelf flat, dry, and empty.
3. Lift the box out of the stow bay and stand it open at the box spot the next episode's config puts it, front right for Config L and M, front left for Config R, square to the front edge.
4. Take the lid off and lay it flat past the box at the lid spot, the right end for Config L and M, the left end for Config R, out of the front lane.
5. Take the four flat packs out of the near half. Leave the near half empty.
6. Shake the four packs out flat and lay them as the new out stack at the spot the next episode's config puts it, front left for Config L, front middle for Config M, front right for Config R, one on another, each squared to the one under it.
7. Fold each of the two rail items once over its own hanger, hook standing up clear of the fold, and lay both in the far half of the box.

8. Fold each of the two shelf items flat and lay both in the far half, under the two on hangers, so the two on hangers are on top.
9. Check the far half now holds four items in the fixed order: two on hangers on top, two folded flat under them.
10. Wipe the bench top and dry it. Leave it clear but for the out stack, the open box, and the lid.
11. Replace the box if a corner is crushed, if the divider is split or loose, or if the lid will not sit down on all four sides.
12. Replace a hanger if its hook is bent, if it is too wide for the rail, or if an item slides off it.
13. Replace an item if it is torn, damp, or so creased that it will not fold to a flat pack.
14. Check the rail is fixed and does not turn or swing, and the shelf is level and does not tilt.
15. Check the stow bay is empty and clear, and wide enough and deep enough for the closed box.
16. Pick up anything that fell on the floor and check the floor is clear and dry.
17. Check the base is still square, centered, and locked, and push it firmly once by hand to confirm it does not roll. If it has moved, park it again by the Base positioning steps.
18. Run the Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the bench layout: the violations below were written for Config L (out stack at the front left taken by the left gripper, box at the front right with the hang items lifted by the right gripper and handed over to the left). The arm-role cues in them will be rewritten later to cover all three configs; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the grippers do does not match the config on the bench. A gripper reaches across the bench for an out item or a hook; a hang item is handed over in Config R,

or is not handed over in Config L or M; the right gripper hangs an item on the rail; the out stack, the box, or the lid is not where that config puts it; or the wrong IF line is followed in Step 1, 2, 3, or 4.

- **SOP rule broken:** Steps 1.1, 3.1, and 4.1, the same-side rule and the hand-over rule. The gripper on the out stack's side takes the out items, the gripper on the box's side lifts the hang items, the left gripper hangs them, the hand-over happens only in Config L and M, and the IF line followed is the one for the config on the bench.
- **Coaching note:** look where the out stack and the box stand before anything is touched, then follow that config's IF lines through Steps 1 to 4.

Violation: Base moved during the episode

- **Visible cue:** the bench shifts in the frame, the bench edge changes angle in the frame, or the base rolls, creeps, or turns at any time after recording starts.
- **SOP rule broken:** Step 8.4 and the Base positioning rule it confirms, the base is parked and locked before recording and does not move at all during the episode.
- **Coaching note:** park it, lock it, push-test it. If the park is wrong, fix it before recording, never during.

Violation: Lifted high or carried over a fixture

- **Visible cue:** an item, the lid, or the box is lifted well above the bench instead of just off what it stands on, or is carried over the rail, over the shelf, or across the back of the bench. The one lift that may go up is the hook going onto the rail.
- **SOP rule broken:** Steps 1.1, 1.4, 3.2, 4.2, 5.2, 6.2, 7.2, and 8.2, lift just off, come forward into the front lane, and carry low and level.
- **Coaching note:** up a little, forward, then along. Only the hook goes up, and only at the rail.

Violation: Carried outside the front lane

- **Visible cue:** a carry goes across the back of the bench instead of along the front lane, or a thing is put down in the lane in the middle of a carry.
- **SOP rule broken:** Steps 1.1, 1.4, 3.2, 4.2, 5.2, 6.2, 7.2, and 8.2, every carry goes along the front lane, and nothing is ever put down in it.
- **Coaching note:** the lane is the road. Nothing crosses the back of the bench, and nothing parks in the lane.

Violation: Leaned on or moved a fixture

- **Visible cue:** a gripper, wrist, or forearm rests on the bench edge, the rail, the shelf, or the bay, the rail turns or swings, the shelf tilts, or the box slides on the bench under a set down.
- **SOP rule broken:** Steps 1 to 8, nothing leans on, pushes, drags, or lifts the bench, the rail, the shelf, or the bay.

- **Coaching note:** the arm holds itself up. If something moves when it is touched, it was touched too hard.

Violation: Dragged instead of lifted

- **Visible cue:** an item, the lid, or the box is slid or dragged along the bench, along the shelf, or along the rail instead of being lifted straight up and carried.
- **SOP rule broken:** Steps 1 to 8, every pick lifts straight up off the surface before it moves anywhere.
- **Coaching note:** lift first, then move. A drag creases the item and pushes the fixture.

Violation: Held by the wrong part

- **Visible cue:** a gripper closes on a sleeve, a hem, or a corner to lift a whole item, closes on the garment of a hang item instead of on the hook, or holds the closed box by the lid or hooks under the lid edge.
- **SOP rule broken:** Steps 1.1, 3.1, 4.1, 5.1, 6.1, and 8.1, take a flat item at the middle of its sides, a hang item by its hook, and the box by its body below the lid.
- **Coaching note:** hook for hangers, sides for packs, body for the box. Never the lid, never a sleeve.

Violation: Took the wrong item off a stack

- **Visible cue:** a gripper pulls an item out from under another one, or lifts two items at once, on the out stack or in the far half of the box.
- **SOP rule broken:** Steps 1.1, 2.1, 2.2, 2.3, 3.1, 4.1, 5.1, and 6.1, the top one is always taken.
- **Coaching note:** top one only. If two come up, put both back and start that pick again.

Violation: Folded off the square or not laid out flat

- **Visible cue:** an item is folded in the air, on the out stack, on the shelf, or inside the box instead of on the fold square, or the first fold is made while a sleeve is trapped, a corner is folded under, or the item is still bunched from the stack.
- **SOP rule broken:** Steps 1.2 and 1.3, the item lies out flat and square on the fold square, sleeves and corners pulled out, before any fold is made.
- **Coaching note:** lay it out first, flat and square, then fold it. A fold over a trapped sleeve never comes out square.

Violation: Wrong number of folds

- **Visible cue:** an item goes into the box after two folds, or after four or more folds.
- **SOP rule broken:** Step 1.3, each item gets three folds: left in, right in, bottom up.
- **Coaching note:** three folds, every item, every time. Count them as they are made.

Violation: Fold not square

- **Visible cue:** the flat pack has edges that run crooked, corners that miss each other by more than a finger's width, or a sleeve, hem, or corner sticking out of the pack.
- **SOP rule broken:** Step 1.3, the pack is a square fold before it leaves the fold square.
- **Coaching note:** line the edge up before bringing it over. Fix it on the square, not in the box.

Violation: Pack put in the wrong half

- **Visible cue:** a folded out item is laid in the far half of the box, or laid across the divider, or laid on top of the in items.
- **SOP rule broken:** Step 1.4, the four flat packs go in the near half and nowhere else.
- **Coaching note:** near half is out, far half is in. Nothing lies across the divider.

Violation: Out items and in items mixed

- **Visible cue:** an in item is left in the near half, a flat pack ends up in the far half, or the two are stacked together at any time.
- **SOP rule broken:** Steps 1.4, 3.1, 4.1, 5.1, and 6.1, the two halves are never mixed.
- **Coaching note:** two halves, two jobs. What comes out never goes back in.

Violation: Done in the wrong order

- **Visible cue:** an in item is taken out of the box before all four out items are folded and in the near half, a shelf item is taken out before both hang items are on the rail, or the lid goes on before the far half is empty.
- **SOP rule broken:** Steps 1 to 7, four out first, then the two hang items, then the two shelf items, then the lid.
- **Coaching note:** fill the box, empty the box, then close it. One thing at a time, in order.

Violation: Item not hanging right

- **Visible cue:** the hook sits on the edge of the rail instead of all the way over it, the hanger hangs crooked, the item hangs off one shoulder, or the hanger swings or slides along the rail when the left gripper lets go.
- **SOP rule broken:** Steps 3.3 and 4.3, the hook goes all the way over the rail and the hanger settles straight before the gripper lets go.
- **Coaching note:** all the way over, then watch it settle before letting go.

Violation: Hangers touching or knocked

- **Visible cue:** the two hangers end up touching, the second hanger pushes the first one along the rail, or the first item is knocked off the rail while the second goes on.
- **SOP rule broken:** Step 4.3, the second hook goes on a hand's width from the first, without touching it.

- **Coaching note:** leave a gap. The rail has room for two, so use it.

Violation: Item not sitting right on the shelf

- **Visible cue:** a folded item hangs over the front edge or a side edge of the shelf, lies crooked, sits on top of the other item, or slides when the grippers let go.
- **SOP rule broken:** Steps 5.3 and 6.3, lay it flat, square, and fully on the shelf, beside the other item.
- **Coaching note:** fully on and square, side by side. Nothing hangs over an edge.

Violation: Pack fell open

- **Visible cue:** a flat pack comes apart in the air, in the box, or on the shelf, so it no longer holds its folded shape.
- **SOP rule broken:** Steps 1.4, 5.1, 5.3, 6.1, and 6.3, lift a pack with a gripper on each side and keep it flat and level.
- **Coaching note:** two sides, level, no swing. A pack lifted from one corner always opens.

Violation: Lid put on wrong

- **Visible cue:** the lid ends up crooked on the box, riding up on a corner, with a gap visible from the front, or resting half on the box and half on the bench.
- **SOP rule broken:** Step 7.3, the lid goes down on all four sides and square with the box.
- **Coaching note:** line it up over the box before lowering it. Straight down, all four sides.

Violation: Box closed too early

- **Visible cue:** the lid goes on while an in item is still in the far half, or while a flat pack is still on the bench or on the fold square.
- **SOP rule broken:** Step 7.1, look in the box and count four packs in the near half and nothing in the far half before the lid goes on.
- **Coaching note:** look in first. A lid on a wrong box hides the mistake.

Violation: Box stowed wrong

- **Visible cue:** the box ends up crooked in the bay, sticking out past the front of the bay, on its side, on the bench, or on the floor.
- **SOP rule broken:** Step 8.3, the box sits flat, square, and fully inside the stow bay.
- **Coaching note:** straight down into the bay, square, and all the way in.

Violation: Wrong arm used

- **Visible cue:** the right gripper folds the left side in or the left gripper folds the right side in, a pack, the lid, or the box is lifted with one gripper only or with the grippers on the wrong sides, or the left gripper flattens a fold.

- **SOP rule broken:** Steps 1.3, 1.4, 5.1, 6.1, 7.2, and 8.1, the left gripper works the left side and the right gripper the right side, the right gripper flattens, and every pack, the lid, and the box are lifted by both grippers together.
- **Coaching note:** left hand left, right hand right, and two hands on anything flat.

Violation: Something dropped, knocked over, or on the floor

- **Visible cue:** an item, a hanger, the lid, or the box slips out of a gripper, falls on the bench or on the floor, is knocked off the rail or off the shelf by an arm going past, or any part of it touches the floor at any time.
- **SOP rule broken:** Steps 1 to 8, one thing moves at a time, each one is put down under control, and clothes go on the bench, in the box, on the rail, or on the shelf, and nowhere else.
- **Coaching note:** close all the way before lifting, and put it down before letting go. A floor touch is reported, not picked back up.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails and the episode carries on with no retry: a sleeve sticks out of a pack, a pack lies across the divider, a hook sits on the edge of the rail, or a folded item hangs over the shelf edge.
- **SOP rule broken:** Steps 1 to 8, do the retry written under each failed check before moving on.
- **Coaching note:** a check is only worth doing if the retry is done. Fix it where it happened.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item out of place, a pack still on the bench, the box open, the box out of the bay, the fold square or the front lane not clear, or an arm away from home.
- **SOP rule broken:** Step 8.4 (confirm the rail, the shelf, the box, the bench, the fixtures, and the locked base; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see inside the box, the rail, or the front edge of the shelf**, so nobody can tell if an item was placed right.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.

- **Base fault:** a brake that releases on its own, or a caster that seizes so the base cannot be parked square.
- **Defective object:** a box with a crushed corner, a split or loose divider, a bottom that folds open, or that will not stand square on the bench; a lid that will not sit down on all four sides on a correctly placed box, or that is the wrong size; a hanger with a bent hook that will not sit over the rail, too wide for the rail, or that lets the item slide off in a correct hang; an item that is torn, damp, stiff, or so creased that it will not fold to a flat pack, or too big to fit the near half. Replace it before the next episode.
- **Fixture fault:** a rail that turns, swings, sags, or comes loose under a correctly placed hook; a shelf that tilts, is too shallow for a folded item to lie fully on, or is wet or dirty at the start; a stow bay too small for the closed box, or blocked by something that was there before the episode.
- **Stacking fault:** the far half arrives in the wrong order, so the two on hangers are not the two on top.
- **Reach fault:** a place turns out to be too far for its arm with the base parked correctly, so the out stack, the fold square, either half of the box, the lid, the rail, the shelf, or the bay cannot be reached without extending the arm to a limit.

Annotation subtasks (from SOP)

1. Take the top out item to the fold square
2. Lay the out item flat on the fold square
3. Fold the out item three times
4. Lay the flat pack in the near half of the box
5. Repeat the fold and box for the second out item
6. Repeat the fold and box for the third out item
7. Repeat the fold and box for the fourth out item
8. Lift the first hang item out of the box by its hook
9. Hand the hang item over to the left gripper (Config L and M only)
10. Carry the first hang item along the front lane to the rail
11. Put the first hook on the rail
12. Lift the second hang item out of the box by its hook
13. Carry the second hang item along the front lane to the rail
14. Put the second hook on the rail
15. Lift the first shelf item out of the box flat
16. Lay the first shelf item on the shelf
17. Lift the second shelf item out of the box flat
18. Lay the second shelf item beside the first

19. Pick up the lid and put it on the box
20. Lift the closed box and stow it in the bay
21. Return both arms home and end the episode